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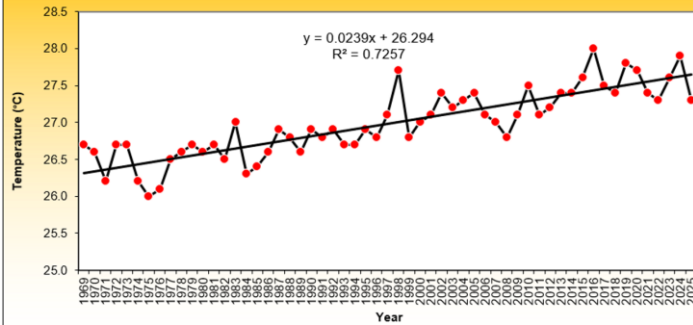
EEEE3054

Renewable Generation Technologies

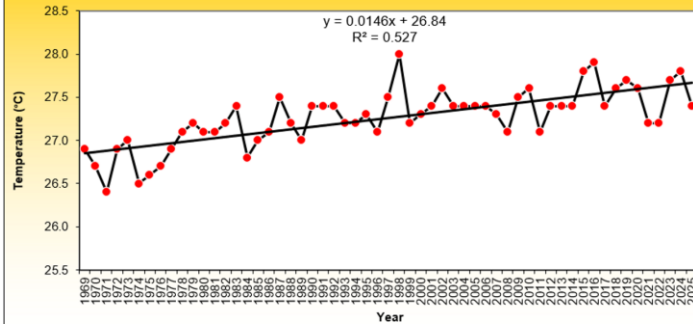
Solar Cells



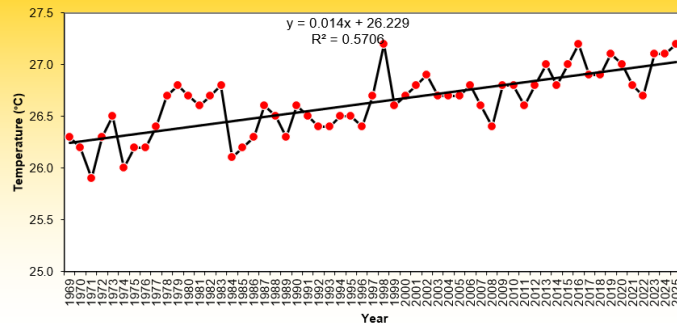
Annual Mean Temperature Trend for Peninsular Malaysia



Annual Mean Temperature Trend for Sabah



Annual Mean Temperature Trend for Sarawak



Temperature Rise in Malaysia

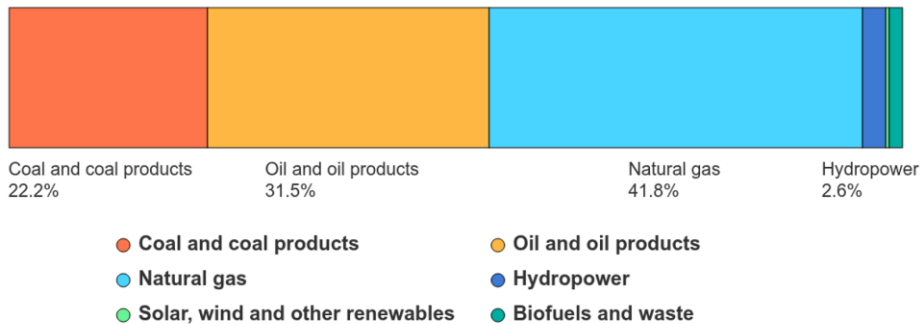
- Rapid economic growth drives **climate change** primarily by increasing greenhouse gas emissions from fossil fuel combustion, urban expansion, and heavy construction.
- Extreme climate change will lead to extreme heat, sea level rise, and sea ice free arctic.
- The temperature rise in Peninsular is the greatest, with about 1°C since 1969.

Source: Malaysian Meteorological Department

<https://www.met.gov.my/en/pendidikan/perubahan-iklim-and-kesan-rumah-hijau/>



Total energy supply, Malaysia, 2023

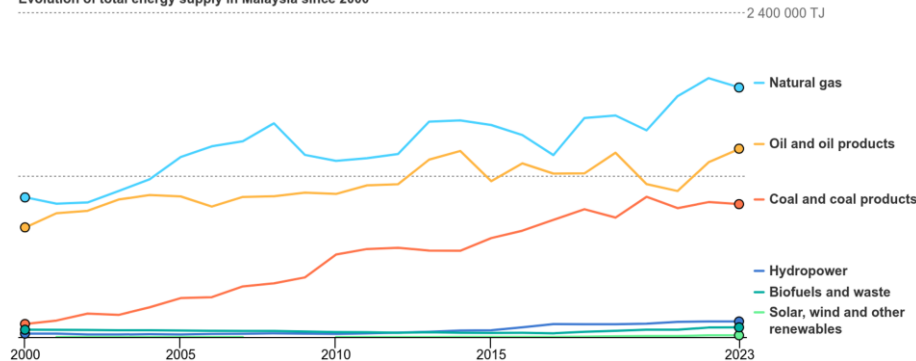


Source: International Energy Agency. Licence: CC BY 4.0

Energy Supply

- Heavily reliant on **fossil fuels**, which account for roughly **95%** of total primary energy.
- Renewable Energy accounts only about **4%** of the energy supply.

Evolution of total energy supply in Malaysia since 2000



Source: International Energy Agency. Licence: CC BY 4.0

Source: Malaysian Meteorological Department
<https://www.iea.org/countries/malaysia/energy-mix>

Climate Change Initiatives

- **Paris Agreement**

- ☐ Limit global warming to well below 2°C, preferably to 1.5°C, compared to pre-industrial levels.

- **The United Nations Framework Convention on Climate Change (UNFCCC)**

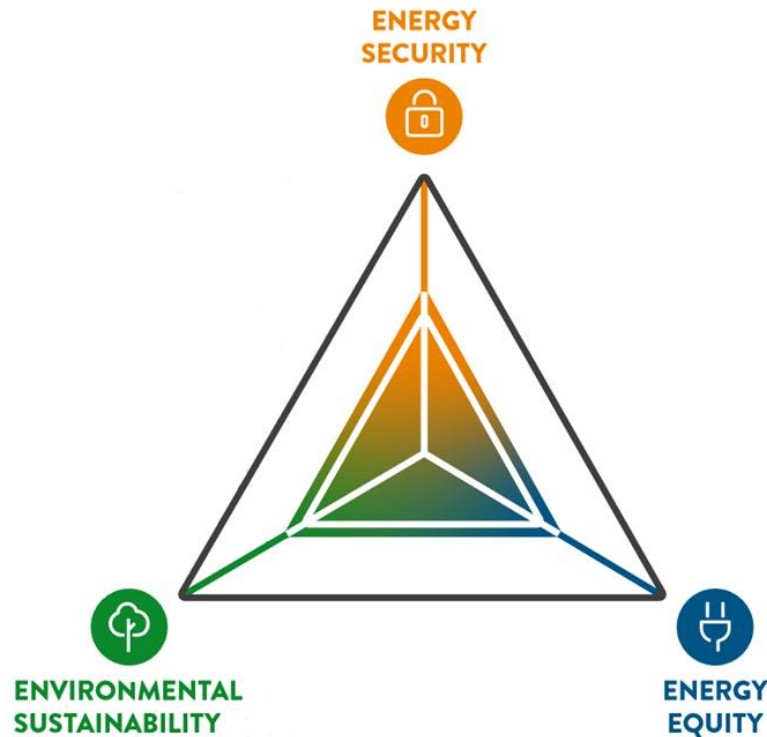
- ☐ SDG 13 - investment in renewable energy



United Nations Framework
Convention on Climate Change



Energy Trilemma



- According to the World Energy Council, the ‘trilemma’ between **Energy Equity**, **Environmental Sustainability** and **Energy Security** is one of the biggest challenges of this transition.
- It is difficult to emphasize one without compromising the other.



Energy Trilemma

- **Energy Equity** of this transition - Ability to provide universal access to affordable, fairly priced and abundant energy for domestic and commercial use.
- **Environmental Sustainability** - Transition of energy system towards mitigating and avoiding potential environment and climate change impacts.
- **Energy Security** - Capacity to meet current and future energy demand reliably, withstand and recover swiftly from system shocks with minimal disruption to supplies.



Timeline

Five-fuel
Policy (2001)

Designed to diversify energy supply by emphasizing **five primary fuels**:

- Oil
- Gas
- Hydro
- Coal
- Renewable energy

National RE
Policy and
Action Plan
(2010)

The **first comprehensive RE policy** framework in Malaysia, outlining objectives, targets and actions for renewable energy deployment, including feed-in tariffs and institutional measures.

National
Energy
Efficiency
Action Plan
(2015)

Focused on **energy efficiency** improvements with quantified targets (e.g., 8% reduction in demand vs baseline)

Malaysia
Renewable
Energy
Roadmap
(2021)

RE roadmap 2021-2035. Sets strategic pathways for RE, including **31% RE share** of national installed capacity by 2025

National
Energy
Policy (2022)

Long-term (2022-2040) policy covering **supply security, sustainability and efficient energy use**

National
Energy
Transition
Roadmap
(2023)

Transition to **low-carbon/ 2050 net-zero**

Energy
Efficiency
and
Conservation
Act (2024)

Translating **energy efficiency policy** into **enforceable law**



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ENERGY EFFICIENCY & CONSERVATION ACT 2024 (EECA 2024)



Energy Efficiency & Conservation Act 2024

- Came into force in 2025 as Malaysia's first dedicated energy efficiency law.
- Introduces mandatory energy efficiency standards for industries, buildings, and large energy users (replaced Efficient Management of Electrical Energy Regulations 2008).



Energy Consumers – Industrial Sectors

- **Industrial facilities (Energy Consumers)** are regulated if their energy use meets or exceeds a threshold of **21,600 gigajoules (GJ) per year**. It is required to:
 - ❑ **Appoint a Registered Energy Manager (REM)** who is responsible for implementing and monitoring energy management systems, preparing energy efficiency reports, and advising on energy-saving measures.
 - ❑ **Establish and operate an Energy Management System (EnMS)** to systematically manage energy consumption.
 - ❑ **Conduct periodic energy audits** to identify inefficiencies and opportunities for savings.
 - ❑ **Submit Annual Energy Efficiency and Conservation Reports** to the Energy Commission, detailing energy consumption, management systems, and proposed improvements.



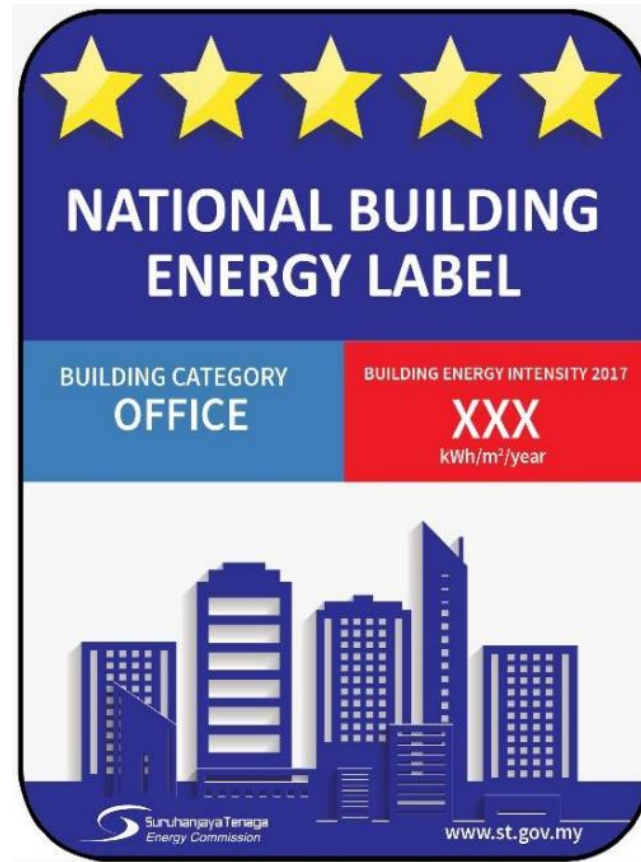
Office Buildings

- **For office buildings and other commercial buildings** that have a gross floor area (GFA) of **8,000 m² (about 86,000 sq ft) or more**:
 - ❑ The Person in Charge (owner or building manager) **must apply for and display a Building Energy Intensity (BEI)** label at a conspicuous part of the building
 - ❑ The energy intensity performance of the building must meet prescribed standards with **at least a two-star rating**. Non-compliance will require the building to undergo an energy audit and implement an energy efficiency improvement plan.
- The BEI is a benchmarking tool to monitor building energy performance by indicating the intensity of energy used per meter square area of the building.
- BEI is calculated by the ratio between annual energy consumption of a building (kWh/year) and building nett floor area (NFA)

$$BEI = \frac{\text{Annual energy consumption (kWh)}}{NFA (m^2)}$$

Building Energy Intensity Label

Building Category:
Office/ Hospital/ University/
School & others



Star rating:

1 star: Not efficient
5 star: Most efficient

**Building energy
performance Unit:**
kWh/m²/year

STAR RATING	BEI RANGE VALUE
5-Star	BEI ≤ 100
4-Star	100 < BEI ≤ 130
3-Star	130 < BEI ≤ 160
2-Star	160 < BEI ≤ 250
1-Star	BEI > 250



Appliance Manufacturers and Importers

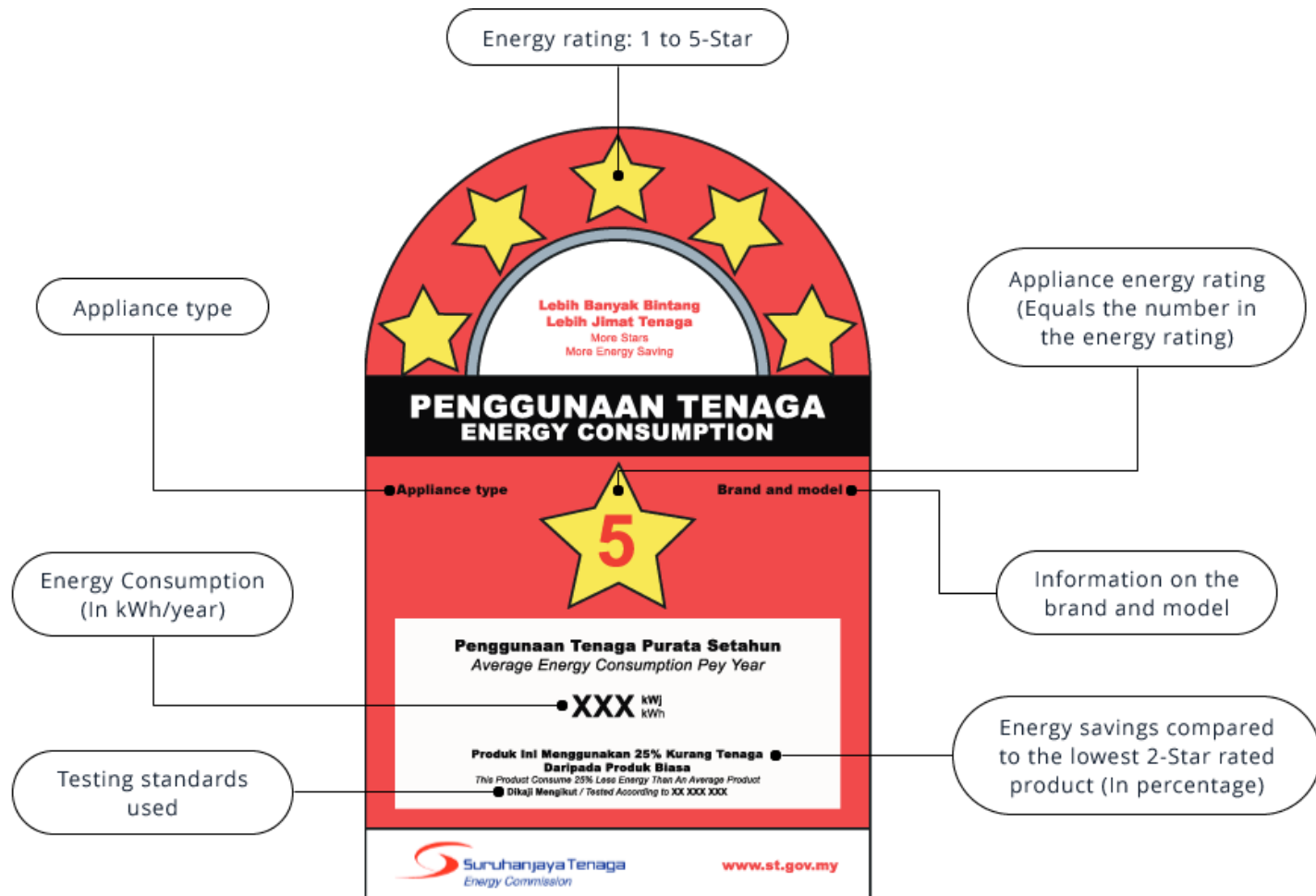
- **Appliance manufacturers and importers** are regulated through product efficiency controls.
- The regulated electrical appliances includes air conditioners, refrigerators & freezers, fans, lighting products, televisions, washing machines, microwave, and others
- EECA requirements are:
 - ❑ Manufacturers and importers are required to obtain a certificate of registration (COR)
 - ❑ Energy using appliances to meet the prescribed minimum requirement and obtain a Certificate of Efficiency (COE).
 - ❑ Energy using appliances to be affixed with an energy efficiency rating label that will also be issued together with COE.



Mandatory Minimum Energy Performance Standards (MEPS)

- Every regulated appliance must meet minimum efficiency levels set by the Energy Commission (ST).
- MEPS minimum requirement is **2-Star**.
- Products that fail MEPS:
 - ☐ Cannot be manufactured for sale
 - ☐ Cannot be imported
 - ☐ Cannot be distributed or sold

Mandatory Minimum Energy Performance Standards (MEPS)





Mandatory Minimum Energy Performance Standards (MEPS)

- Air Conditioner
 - ☐ Type :Non-ducted Single Split Wall Mounted
 - ☐ Capacity $\leq$ 25,000 btu/hr
- Fan
 - ☐ Ceiling Fan with diameter less than 60 inch
 - ☐ Wall fan,desk fan,table fan with diameter less than 16 inch
- Refrigerator
 - ☐ 1-door & 2-door only
- Television
 - ☐ Type :LCD,PLASMA,LED,CRT
 - ☐ Screen size up to or equal to 70 inch
- Lamp
 - ☐ T5 & T8 Fluorescent Lamp
 - ☐ Self ballasted single capped CFL
 - ☐ Single Capped Fluorescent Lamp & Circular Fluorescent Lamp
 - ☐ Self ballasted LED Lamp

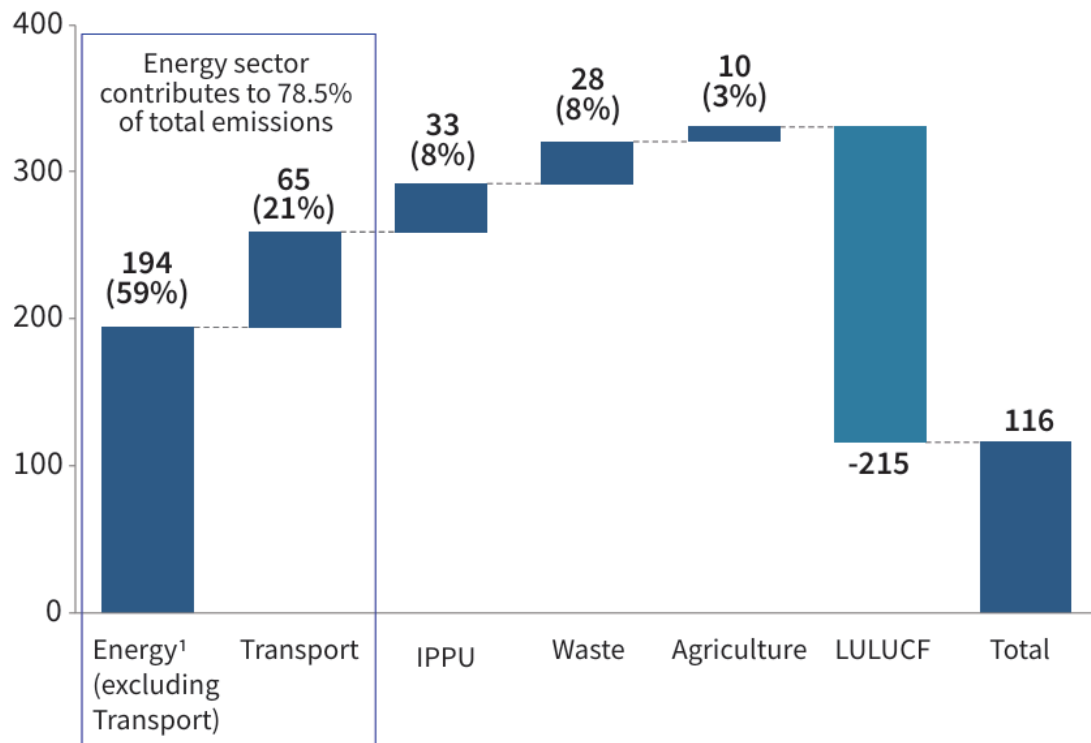


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ENERGY POLICIES ON RENEWABLE ENERGY

Malaysia Greenhouse Gas (GHG) Emission



In 2019, the **energy sector** emitted 259,326.11 Gigagram CO₂ equivalent (GgCO₂eq), or 78.5% of total emissions. This was followed by industrial processes and product use (IPPU) at 10%, waste at 9% and agriculture at 3%.

1 Refers to emissions from energy industries, manufacturing industries and construction, other sectors and non-specified energy emissions, and fugitive emissions from fuels.



National Energy Policy (NEP)

- NEP provides a long-term, integrated direction for Malaysia's **energy sector** up to 2040, while supporting the country's energy transition and climate commitments.
- The policy aligns energy planning with economic development, environmental sustainability, and social well-being, and serves as the umbrella under which roadmaps such as **MyRER** and **NETR** operate.
- NEP has three main objectives (balance energy trilemma):
 - enhancing macroeconomic resilience and energy security
 - achieving social equitability and affordability
 - ensuring environmental sustainability



National Energy Policy (NEP)

Selected Targets			2018	Low Carbon Nation Aspiration 2040
	1. Percentage of urban public transport modal share	● ●	20%	50%
	2. Percentage of electric vehicle (EV) share	●	<1%	38%
	3. Alternative fuel standard for heavy transport	●	B5	B30
	4. Percentage of Liquefied Natural Gas (LNG) as alternative fuel for marine transport	●	0%	25%
	5. Percentage of industrial and commercial energy efficiency savings	● ● ●	<1%	11%
	6. Percentage of residential energy efficiency savings	● ● ●	<1%	10%
	7. Total installed capacity of RE	● ● ●	7,597 MW	18,431 MW
	8. Percentage of coal in installed capacity	● ● ●	31.4%	18.6%
	9. Percentage of RE in TPES	● ● ●	7.2%	17%



National Energy Transition Roadmap (NETR)

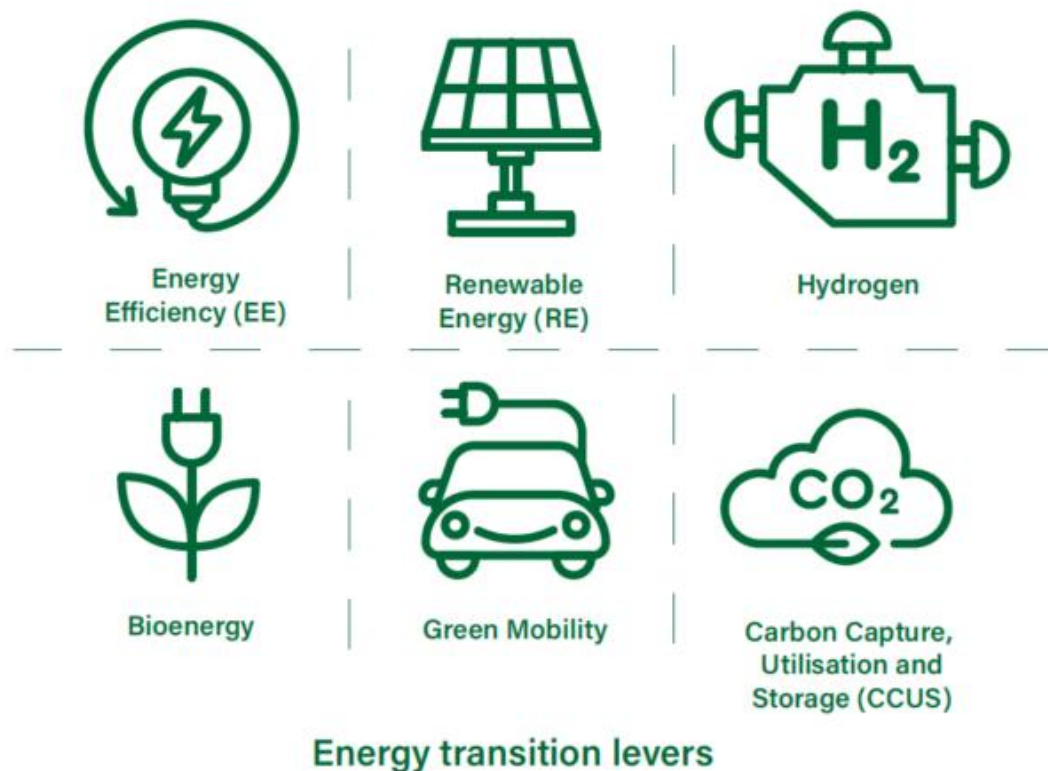
- NETR is national roadmap to achieve net-zero GHG emission by 2050.
- It provides strategic plan to shift energy systems from fossil fuel-based to greener and low-carbon systems.
- NETR aims to lower the energy sector's greenhouse gas emissions by **32% by 2050**, relative to the 2019 baseline.



National Energy Transition Roadmap (NETR)

NETR identifies 10 flagship catalyst projects under six key levers, including

- ☐ Energy efficiency
- ☐ Renewable energy
- ☐ Hydrogen
- ☐ Bioenergy
- ☐ Green mobility
- ☐ Carbon capture, utilization and storage



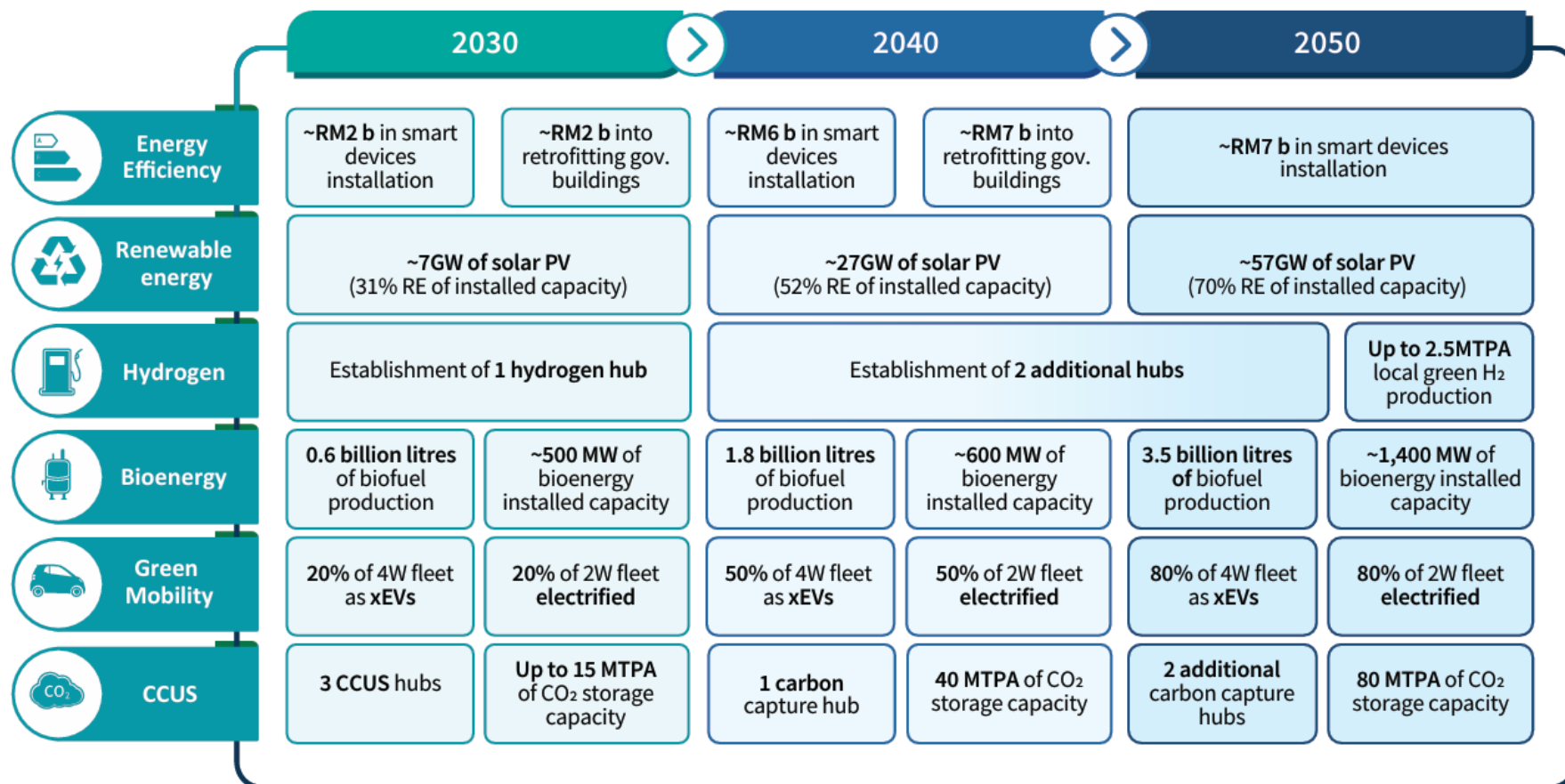


Overview of NEP and NETR Goals

Sector and Key Driver		2040 DTN Low Carbon Nation Asp.	2050 NETR Responsible Transition
EE Energy Efficiency	Industry and Commercial energy efficiency savings (%)	11%	23%
	Residential energy efficiency savings (%)	10%	20%
RE Renewable energy	Coal share of installed capacity (%)	19%	0%
	RE share of installed capacity (%)	41%	70%
HY Hydrogen	Green hydrogen production (MTPA)	N/A	Up to 2.5 MTPA
	Grey hydrogen feedstock phase off (%)	N/A	100%
	Hydrogen hubs (#)	N/A	3
BI Bioenergy	Biofuel capacity (billion litres)	N/A	3.5
	Bioenergy power generation (GW)	N/A	1.4
GM Green Mobility	Land		
	Urban public transport modal share (%)	50%	60%
	xEV (4W) share of fleet (%)	38%	80%
	E2W share of fleet (%)	N/A	80%
	Light vehicle fuel economy	N/A	~30%
	Heavy transport fuel economy	N/A	~24%
	Biofuel blending for heavy transport (%)	B30	B30
	Hydrogen penetration for heavy transport (%)	N/A	5%
	Marine		
	LNG penetration as alternative fuel in marine transport (%)	25%	N/A
Av.	Green fuel penetration in marine transport (%)	N/A	40%
	SAF blending mandate by 2050 (%)	N/A	47%
CC CCUS	Number of CCUS clusters (#)	N/A	3-6
	CO ₂ storage capacity (Mtpa)	N/A	40-80



National Energy Transition Roadmap (NETR)





Malaysia Renewable Energy Roadmap (MyRER)

- The primary goal of MyRER is to increase the share of renewable energy in Malaysia's power capacity mix. The specific targets are:
 - ▣ **31% share of RE** in the national installed capacity mix by **2025**.
 - ▣ **40% share of RE** in the national installed capacity mix by **2035**.
 - ▣ **70% share of RE** in the national installed capacity mix by **2050 (NETR)**.



31%
2025

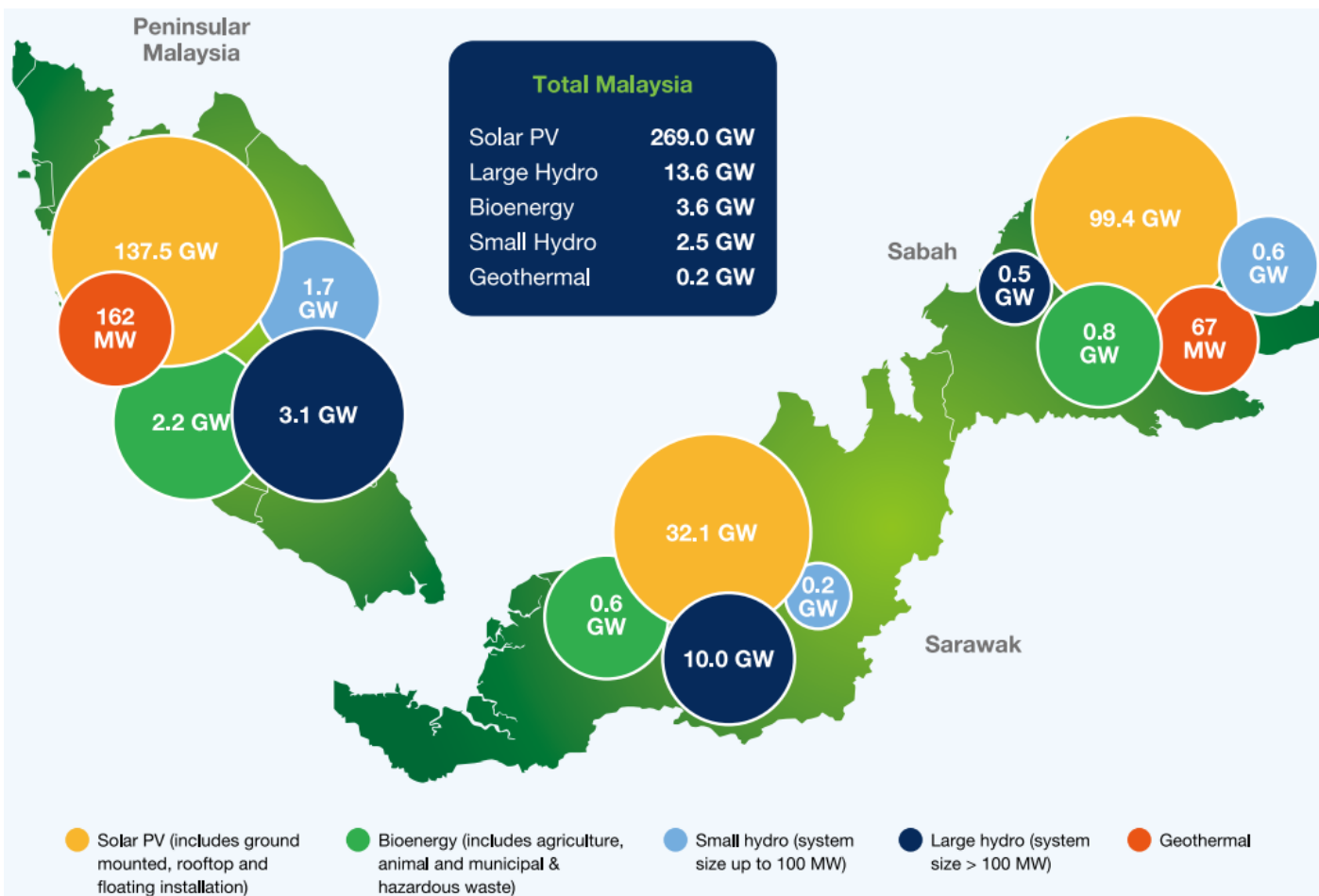
40%
2035

70%
2050



Malaysia Renewable Energy Roadmap (MyRER)

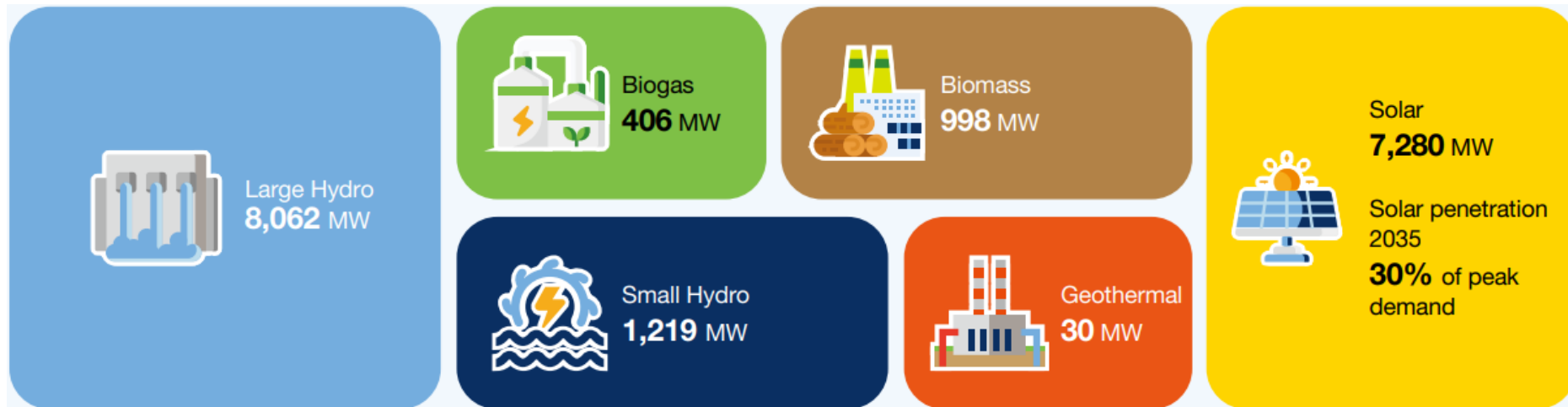
Malaysia has high potential for RE, due to **year-round solar irradiation**, **Abundant biomass resources**, **Extensive river basins**





Malaysia Renewable Energy Roadmap (MyRER)

- This represents a projected increase in RE capacity of **17,996 MW** by **2035**.



RE Capacity target in 2035

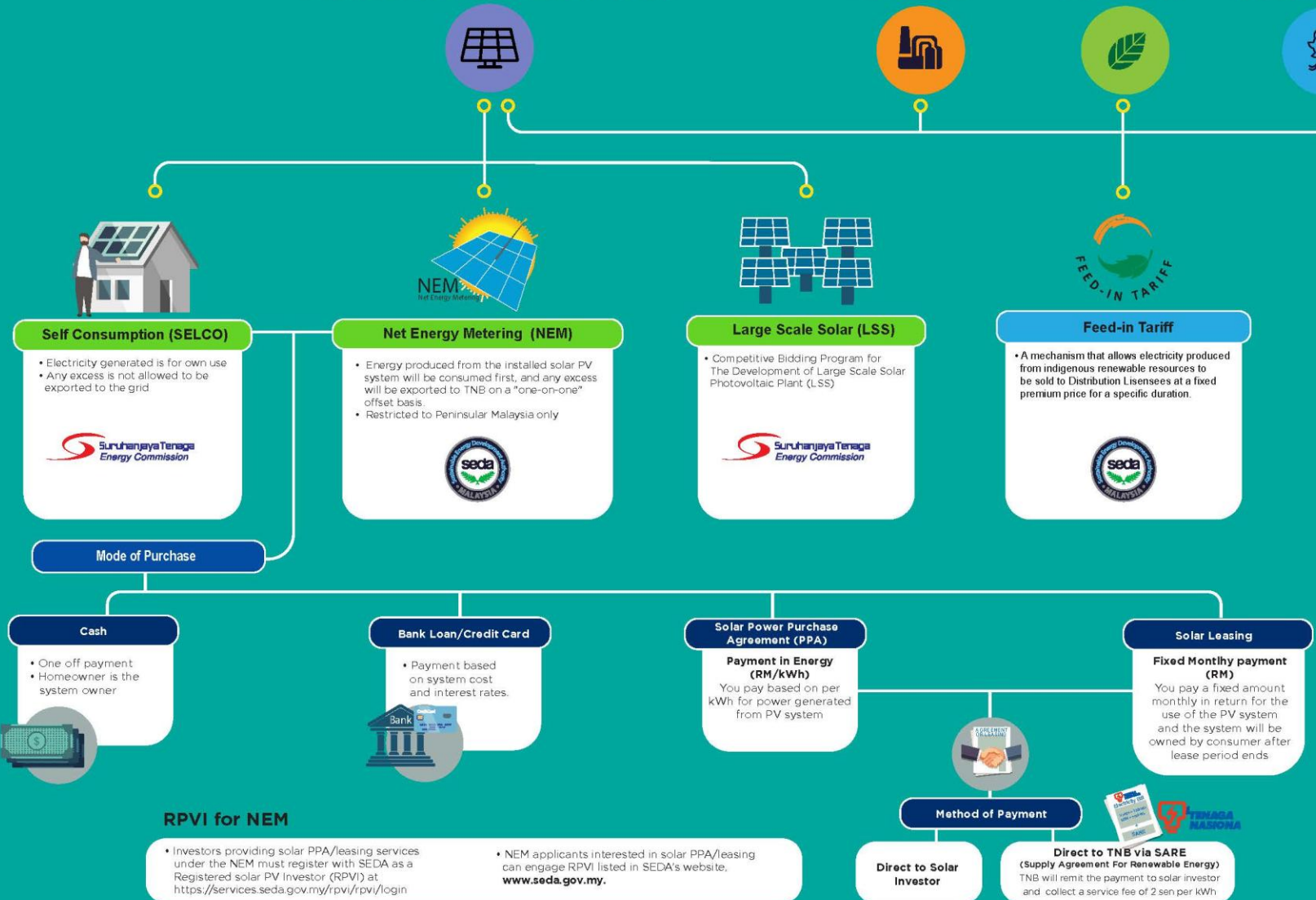
RENEWABLE ENERGY PROGRAMMES IN MALAYSIA

SOLAR PHOTOVOLTAIC (PV)

BIOGAS

BIOMASS

SMALL HYDRO





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QUESTIONS?